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Epilepsia

BRIEF COMMUNICATION

Automated detection of tonic seizures using wearable movement sensor and artificial neural network

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Abstract

Although several validated wearable devices are available for detection of generalized tonic–clonic seizures, automated detection of tonic seizures is still a challenge. In this phase 1 study, we report development and validation of an artificial neural network (ANN) model for automated detection of tonic seizures with visible clinical manifestation using a wearable wristband movement sensor (accelerometer and gyroscope). The dataset prospectively recorded for this study included 70 tonic seizures from 15 patients (seven males, age 3–46 years, median = 19 years). We trained an ANN model to detect tonic seizures. The independent test dataset comprised nocturnal recordings, including 10 tonic seizures from three patients and additional (distractor) data from three subjects without seizures. The ANN model detected nocturnal tonic seizures with visible clinical manifestation with a sensitivity of 100% (95% confidence interval = 69%–100%) and with an average false alarm rate of .16/night. The mean detection latency was 14.1 s (median = 10 s), with a maximum of 47 s. These data suggest that nocturnal tonic seizures can be reliably detected with movement sensors using ANN. Large-scale, multicenter prospective (phase 3) trials are needed to provide compelling evidence for the clinical utility of this device and detection algorithm.

KEYWORDS

artificial intelligence, automated seizure detection, tonic seizures, wristband sensor

1 | INTRODUCTION

There is need for automated detection of epileptic seizures using wearable devices, for safety indications and for objective seizure quantification.[1] A recently published clinical practice guideline of the International League Against Epilepsy and the International Federation of Clinical Neurophysiology (IFCN) found evidence for reliability of wearable devices to detect generalized tonic–clonic seizures (GTCS) including focal-to-bilateral tonic–clonic seizures, and emphasized the need for further research and development to detect other seizure types.[1]

Tonic seizures typically occur in patients with Lennox–Gastaut syndrome (LGS).[2] Their automated detection is important for clinical decision-making and for drug trials, because the seizure diaries kept by parents are unreliable.[3,4] Furthermore, automated detection of tonic seizures is needed also for safety reasons, because these seizures often occur during the night in unsupervised patients.[2]

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Relatively few studies have addressed automated detection of tonic seizures, reporting modest performance, not sufficient for large-scale clinical implementation. In a phase 1 study using surface electromyography (sEMG) and machine learning (random forest classifier), high sensitivity (100%) was achieved with patient-specific (individualized) parameters, but at the cost of high false alarm rate (up to 190/day).[5] Visual inspection of quantitative sEMG plots was able to identify tonic seizures, but the automated algorithm targeted the tonic–clonic seizures.[6] Using a multimodal wearable device (accelerometry and heart rate) for nocturnal monitoring in a phase 3 study, an overall seizure detection sensitivity of 77% was achieved for major motor seizures other than tonic–clonic seizures, including tonic seizures lasting longer than 30 s.[7] A recently published phase 4 study using the same device found an overall sensitivity of 53% for tonic seizures.[8] Automated video analysis using artificial intelligence identified tonic seizures with a low sensitivity (13%–50%) in a phase 2 and a phase 3 study.[9,10] Offline manual detection of tonic seizures using a multimodal wearable device (electroencephalography [EEG], EMG, accelerometry, and gyroscope) had an overall sensitivity of 41%, with a median of 18 false alarms per day.[11] In a phase 1 study, a wrist-worn smartwatch detected tonic seizures with a sensitivity of 77.8% and a specificity of 60%,[12] and another accelerometer device detected all three tonic seizures, which occurred in a series of mixed motor seizures.[13] A nocturnal, heart rate-based seizure detection algorithm that automatically adapts to the patient detected six of 13 tonic seizures (sensitivity = 46%).[14] A movement sensor embedded into the mattress detected three of 11 tonic seizures (sensitivity = 27%).[15] Thus, there is need for much improvement in this area.

The gold standard for detecting epileptic seizures is video-EEG (vEEG) including sEMG, with an added diagnostic value for detecting and classifying seizures with motoric components. vEEG and sEMG in a full setup are not suitable for application outside the hospital as a wearable device. The currently available wearable automatic seizure detection devices are validated for detecting generalized tonic–clonic seizures. Here, we report development and testing of an artificial neural network (ANN) model for automated detection of tonic seizures using a wearable movement sensor (accelerometer and gyroscope) in a phase 1 clinical validation study.[16]

2 | MATERIALS AND METHODS

Thirty-three consecutive patients (16 male, age = 3–75 years, median = 19 years), at risk of having tonic seizures, admitted to long-term vEEG monitoring for diagnostic reasons, were prospectively recruited at the Danish Epilepsy Center. The study was approved by the ethics committee of Region Zealand (SJ-693), and patients or caregivers gave their written informed consent. The patients were not confined to bed during the monitoring but were able to move freely in their own bedroom and interact with other patients in the living room and dining room.[17] The patients were encouraged to use a treadmill or cycle ergometer or play video games involving motor activities during the monitoring. Exclusion criteria for our study included the absence of recorded tonic seizures.

The gold standard for identifying seizures was clinical expert evaluation of long-term vEEG recordings, including polygraphic channels. The recording array comprised EEG (19–25 electrodes for diagnostic monitoring), electrocardiographic, and sEMG electrodes, placed as specified in the guideline of the IFCN.[18] Besides the standard clinical setting, participants wore a wristband movement sensor device. Two experts evaluated the vEEG recordings blinded to the data from the wearable device and annotated the start and stop time of the tonic seizures. A total of 18 patients had recorded seizures. Patients were monitored for 1–3 days (median = 2 days). Movement sensor data from an additional three subjects without seizures were included, to increase the amount of distractor (non-seizure) data. Eight patients were excluded due to technical errors on the seizure detection device (failed recording due to depleted battery or error with the continuous data streaming). This amounts to 25.85% of the total monitoring time. Seven patients did not have tonic seizures. All recorded data were analyzed.

The data used in this study for automated seizure detection were collected using a wrist-worn device with an accelerometer and a gyroscope, developed by Danish Care Technology. The data consist of six continuous axis measurements (three axis accelerometer and three axis gyroscope). Based on our previous experience with the movement sensor,[19] a minimum acceleration of .1 g was required as a threshold for starting the feature extraction. When the threshold was exceeded, features were extracted in 10-s sliding windows, with 1-s overlap. For each window, 594 features were automatically extracted, consisting of statistical measures such as mean, variance, and SD for each of the six sensor readings.

The 18 patients with tonic seizures had a total of 80 seizures, where 10 were discrete seizures identifiable only on sEMG but without visible movement (below the threshold), resulting in 70 recorded clinical seizures from 15 patients (seven males, age = 3–46 years, median = 19 years). The data were split 80%/20% for training and test. The total duration of the training dataset was 123.3 h. After the development of the algorithm, two further patients with seizures were included in the

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study and used as an independent test dataset. The development dataset included 57 tonic seizures (46 for training and 11 for validation), and the independent test dataset included 10 tonic seizures. To assess the specificity (false alarm rate), besides the patient data mentioned above (recorded in the hospital), we included nocturnal data from home recordings of three additional healthy control subjects (i.e., certainly without seizures) age-matched with the patients in the independent test dataset. Hence, the independent test dataset consisted of 2560 min (43 h) of nocturnal recordings, including 10 tonic seizures from three patients and nocturnal data from three subjects without seizures. Three of the six patients in the test dataset were males, and the median age was 43.5 years (range = 16–75 years).

Because epileptic seizures are rare events, there was a significant imbalance between the numbers of time epochs with versus without tonic seizures (the time without seizures being much longer than the time with seizures). To address this issue, for the development of the algorithm we employed data augmentation techniques such as class weighting and oversampling to artificially increase the number of seizure samples. The oversampling only works if the seizure lasts longer than 10 s. If, for example, a seizure lasts 30 s, it is classified as three seizures of 10-s duration.

An ANN model (Figure 1) was built using the Keras API of TensorFlow. It is a sequential model with several layers, including GaussianNoise, Dropout, Dense, and Activation layers. The input data shape is defined by the number of extracted features (594), and GaussianNoise is applied to the layers to add noise to the data. Dropout layers are also added to randomly remove a percentage of the neurons to help prevent overfitting. There are two densely connected layers with 192 neurons in each using the rectified linear unit and scaled exponential linear unit, respectively, as the activation functions. Finally, a dense output layer with a sigmoid activation function is added to produce a probability of a seizure; a score higher than or equal to .5 was chosen as the activation for this final layer.

The model is compiled using the Adam optimizer, with a learning rate of .001 and binary cross-entropy loss function. During the development, we used area under the precision-recall curve (AUC-PR) as performance metric. Early stopping was implemented using AUC-PR to prevent overfitting. The developed model was generic (not personalized).

For clinical validation, we used the independent test dataset (not used for development) including nocturnal recordings, and we evaluated the performance of the automated detector of tonic seizures by determining the sensitivity and false alarm rate per night.

3 | RESULTS

Demographic and clinical data of all included patients are detailed in Data S1. The algorithm automatically extracted >600 features, with the temporal acceleration feature identified as the most significant (Data S2). In the independent test dataset, automated seizure detection of tonic seizures with visible clinical manifestation, using the wrist-worn movement sensor and ANN,

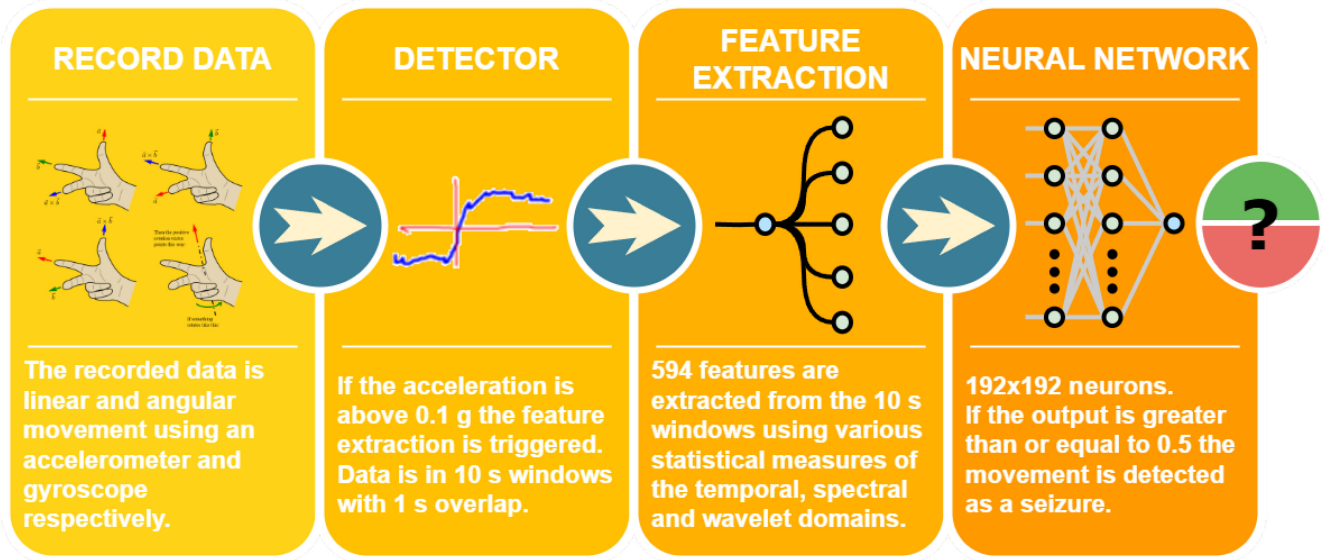

FIGURE 1 Flowchart of automated data analysis, showing the key sequential steps: data recording, detection of input signals, feature extraction, and automated classification using the artificial neural network.

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had a sensitivity of 100% (95% confidence interval [CI] = 69%–100%), with all seizures detected. The false alarm rate was .023/h in the independent test dataset. The average sensitivity in the entire dataset (80 seizures) was 96% (95% CI = 89%–99%), with .23 false alarms per night. In the entire dataset recorded from the 18 patients, the median false alarm rate per patient was 0/night (interquartile range = 0–1, 95% CI = .44–.88/night; Data S1). False alarms were triggered by voluntary movements, arousals, and physiological sleep movements/stretching the limbs. The mean detection latency was 14.1 s (median = 10 s), with a maximum of 47 s. The shortest detection delay was 10 s due to the size of the analysis time window.

4 | DISCUSSION

Overall, the ANN model developed in this study shows promising results in detecting tonic seizures using accelerometer and gyroscope data from a wristband device. All seizures with visible clinical manifestation were detected, with one false alarm per six nights. Compared to recent studies on detecting tonic seizures, we show a very high sensitivity and a low and acceptable false alarm rate,[5–15,20] but the relatively low number of patients and recorded seizures in the independent test dataset is the main limitation of this study, which therefore qualifies as phase 1.

We aimed at detecting nocturnal tonic seizures, because most patients are unsupervised during the night, and many tonic seizures in LGS typically occur during the night. It has earlier been shown that the sensitivity of seizure diaries decreases during nighttime, and the sensitivity of an automatic detection device increased at the same time.[11] Furthermore, prolonged tonic seizures are associated with postictal generalized EEG suppression, which is a reported risk factor for sudden unexpected death in epilepsy (SUDEP).[21,22] The risk of SUDEP and other seizure-related injuries combined with the unreliability of seizure diaries demonstrates the importance of an objective measure for tonic seizures to optimize treatment, and in the future optimally protect patients against seizure-related injuries by alarming caregivers when a seizure occurs. Although the intended use of this algorithm is detection of tonic seizures, it has the potential to detect tonic–clonic seizures too. We recorded three tonic–clonic seizures, and all were detected by the algorithm. A seizure detection algorithm for tonic–clonic seizures, which has been validated in a phase 3 study,[19] is available on a wristband sensor. By running both algorithms concurrently, it becomes feasible to cover both seizure types using the same sensor. We excluded subtle tonic seizures without observable semiology. Although sEMG is important for diagnosing/characterizing tonic seizures, the significance of quantifying subtle tonic seizures, which exhibit sEMG activity without causing limb movements (clinical manifestation), is debatable.

The algorithm in our study has not yet been implemented in the wristband device, and therefore the data processing was done offline on a PC. It may be possible to decrease the number of features by more thoroughly examining the importance of each feature, thus making the algorithm less computationally demanding. With recent advancements within embedded electronics, it is feasible to implement the algorithm in a wrist-worn device in the future for real-time automatic detection of tonic seizures. An offline version of the seizure detection model can be used to objectively detect the number of nocturnal seizures, and on that background preventive measures can be implemented by physicians and caregivers.

The algorithm reported in this paper targets tonic seizures only. To detect other major motor seizure types (such as GTCS) in patients with LGS, it is necessary to run in parallel other algorithms on the same sensor, targeting GTCS. Although the accuracy of those algorithms has previously been reported,[19] it is likely that running both algorithms in parallel will result in a higher false alarm rate.

Large-scale, multicenter prospective (phase 3) studies, with real-time seizure detection using the model reported in this paper, are needed to provide compelling evidence for its clinical utility. This phase 1 study showed promising results and warrants further development and research using this model.

AUTHOR CONTRIBUTIONS

Sidsel Armand Larsen: Study design; formal analysis; investigation; methodology; visualization; writing–original draft preparation. **Daniel Højrup Johansen:** Formal analysis; investigation; methodology; writing–editing and review. **Sándor Beniczky:** Conceptualization; formal analysis; visualization; writing–original draft preparation.

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CONFLICT OF INTEREST STATEMENT

D.H.J. is an employee of Danish Care Technology, a company that produces wearable wristband devices. The company supplied the wearable devices for the study. We confirm that we have read the Journal's position on issues involved in ethical publication and affirm that this report is consistent with those guidelines.

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DATA AVAILABILITY STATEMENT

Deidentified data that support the findings of this study are available upon reasonable request, approval pending intent of use.

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SUPPORTING INFORMATION

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